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A Simple Monte Carlo Method for Estimating Power in Multilevel Designs

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Abstract

Estimating power for multilevel models is complex because there are many moving parts, several sources of variation to consider, and unique sample sizes at Level 1 and Level 2. Monte Carlo computer simulation is a flexible tool that has received considerable attention in the literature. However, much of the work to date has focused on very simple models with one predictor at each level and one cross-level interaction effect, and approaches that do not share this limitation require users to specify a large set of population parameters. The goal of this tutorial is to describe a flexible Monte Carlo approach that accommodates a broad class of multilevel regression models with continuous outcomes. Our tutorial makes three important contributions. First, it allows any number of within-cluster effects, between-cluster effects, covariate effects at either level, cross-level interactions, and random coefficients. Moreover, we do not assume orthogonal effects, and predictors can correlate at either level. Second, our approach accommodates models with multiple interaction effects, and it does so with exact expressions for the variances and covariances of product random variables. Finally, our strategy for deriving hypothetical population parameters does not require pilot or comparable data. Instead, we use intuitive variance-explained effect size expressions to reverse-engineer solutions for the regression coefficients and variance components. We describe a new R package `mlmpower` that computes these solutions and automates the process of generating artificial data sets and summarizing the simulation results. The online supplemental materials provide detailed vignettes that annotate the R scripts and resulting output.

Translational Abstract

A power analysis is a foundational research activity that involves determining the minimum sample size required to detect a presumed effect. One such method for estimating power, Monte Carlo computer simulation, determines the likelihood of detecting an effect by generating a large number of artificial data sets from a hypothetical population. This tutorial article outlines a novel Monte Carlo power estimation method for multilevel regression models. Multilevel data are exceedingly common throughout psychology and the behavioral sciences; examples of such nested data structures include children within classrooms, individuals within families, employees within workgroups, and repeated measurements within individuals, to name a few. Monte Carlo methods for estimating power in multilevel designs are underdeveloped and generally limited to very simple cases that are not representative of published applications. Our method allows researchers to estimate power and determine sample for a broad range of realistic applications. We provide R software functions that automate the process of generating artificial data sets and summarizing the simulation results, and we provide detailed vignettes that annotate the R scripts and resulting output.

Keywords: multilevel models, power, sample size determination

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Power analyses are a foundational research activity, codified in the behavioral sciences by Cohen's (1962, 1977, 1988) seminal work and the countless commentaries, tutorials, and books that followed. Power estimation methods are well established for many common analytic frameworks, particularly analysis of variance (ANOVA) and multiple regression, and software tools abound

(e.g., Champely, 2020; Faul et al., 2007; Muthén & Muthén, 2002; Zhang & Yuan, 2018). The same is not true for multilevel regression models, where user-friendly power analytic methods for realistically complex models are sorely needed. Our tutorial fills this gap, illustrating a simple Monte Carlo simulation approach that accommodates a broad range of multilevel regression models that typify real-world applications.

Power analyses for multilevel designs require unique considerations that make them more challenging than those for single-level regression models. For one, there are two sample sizes to consider—the number of Level 2 units (clusters) and the number of Level 1 observations within each cluster—and pulling these two levers does not have a uniform effect on power. Moreover, the optimal blend of the two sample sizes varies across parameters; some parameters are sensitive to changes in the within-cluster sample

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size, while others are mostly sensitive to changes in the number of Level 2 units (Mathieu et al., 2012; Spybrook et al., 2016). The use of covariates can further moderate the interplay of the two sample sizes (S. W. Raudenbush, 1997; Reise & Duan, 2003).

The proportion of between-cluster variance in the outcome (the intraclass correlation) is also a unique and strong determinant of power in multilevel designs. The magnitude of the intraclass correlation can vary dramatically across research contexts, ranging from very low values in cross-sectional designs to very high values in repeated measures studies. Previous work suggests that larger intraclass correlations can improve the power to detect between-cluster effects, whereas smaller pots of group-level variation are favorable for within-cluster effects (Mathieu et al., 2012; S. W. Raudenbush & Liu, 2000; Scherbaum & Pesner, 2019).

The literature describes three approaches to multilevel power estimation: formula-based strategies that use analytic standard error expressions, a new summary statistics-based method that converts a multilevel power problem to a single-level one, and Monte Carlo computer simulations. We briefly describe these three approaches and point interested readers to Table 2 in Murayama et al. (2022) for an excellent summary of these approaches along with their requisite assumptions and inputs.

Analytic approximations to power have a long history in the multilevel literature (e.g., T. A. Snijders & Bosker, 1993). The general idea is to derive an approximate sampling variance (squared standard error) for a coefficient that features the Level 1 and Level 2 sample sizes and the intraclass correlation (or variance components) as inputs. Given a presumed effect size, this variance expression is then converted to a standardized test statistic, and the area under a noncentral distribution function determines power. Approximations have been developed for a variety of specific contexts, including two-group comparisons (Moerbeek et al., 2000), cluster-randomized designs (S. W. Raudenbush, 1997), multisite trials (S. W. Raudenbush & Liu, 2000), multilevel logistic regression (Moerbeek & Maas, 2005), longitudinal designs with missing data (Hedeker et al., 1999), and three-level models with cross-level interaction effects (Spybrook et al., 2016), among others.

A recent article by Murayama et al. (2022) described a power analysis approach that leverages analytic approximations, albeit in a very different way. The main thesis of their work is that, under certain circumstances, test statistics from multilevel models are equivalent to those obtained from single-level analyses based on aggregated data. The analytic equivalencies in their article allow researchers to use a test statistic from a published multilevel study to formulate a power analysis with conventional software like G*Power (Faul et al., 2007). This so-called summary statistic approach is most applicable when the effect of interest is at Level 2 and the planned study will have the same within-cluster sample size as the study contributing to the test statistic. More complex applications are possible, but the necessary inputs may not be readily available from published studies.

Monte Carlo computer simulation is a third power analytic method that has received considerable attention in the literature (Arend & Schäfer, 2019; Arnold et al., 2011; Bolger & Laurenceau, 2013; Browne et al., 2009; Green & Macleod, 2016; Lافit et al., 2021; Lane & Hennes, 2018; Maas & Hox, 2005; Mathieu et al., 2012). The first step of the procedure specifies hypothetical population parameters. This is arguably the most difficult part of the process because multilevel models have many moving parts and multiple sources of explained variance. The next step uses statistical software to “sample” many artificial data sets from the population and fit the

target model of interest to each. When the population effect size is nonzero, the proportion of data sets that produced a significant test statistic provides a simulation-based power estimate.

Monte Carlo methods are often praised for their flexibility, yet this flexibility is precisely what makes them difficult to implement in multilevel designs. We believe there is substantial room for improving and refining the methodology. For example, much of the work to date has focused on very simple models with one predictor at each level and one cross-level interaction effect (Arend & Schäfer, 2019; Lافit et al., 2021; Maas & Hox, 2005; Mathieu et al., 2012). Limiting the number of variables in a power simulation may preclude the consideration of covariates, which are potentially important determinants of sample size allocation in multilevel designs (Scherbaum & Ferreter, 2009; Spybrook et al., 2011). Additionally, the literature has mostly focused on “pure” within-cluster (i.e., group mean-centered) predictors that are orthogonal to both the Level 2 moderator and cross-level interaction. Few studies have considered Level 1 predictors with nonzero intraclass correlations (e.g., Mathieu et al., 2012), and those that have assume that the regressor’s group means are orthogonal to other predictors.

Power estimation for interaction effects is another area that warrants additional work. The variances and covariances of the predictor variables figure prominently in proportion of explained variance effect size expressions, yet studies have thus far failed to consider that a product variable’s parameters (its expected value, variance, and covariances with other variables) are fully determined by the lower-order moments of its component regressors (Bohrnstedt & Goldberger, 1969). Instead, studies often apply standardized effect sizes that treat interaction terms just like any other predictor variable (e.g., Arend & Schäfer, 2019; Maas & Hox, 2005; Mathieu et al., 2012). While this approach is sometimes justified in simple models with orthogonal effects and standardized variables, it does not generalize to more complex models.

A third avenue for improvement concerns the vital initial step of specifying realistic population parameters. A number of existing approaches to estimating power require researchers to specify presumed values for the population parameters (Bolger & Laurenceau, 2013; Bolger et al., 2012; Browne et al., 2009; Lافit et al., 2021; Lane & Hennes, 2018). Given the sheer number of moving parts in a multilevel model, this is a daunting task, even for very simple analyses. The authors usually address this challenge using estimates from a real data analysis as surrogates for the population values. However, this approach presupposes that researchers have access to a data set with the same structural features and effects as the planned analysis. Absent pilot data, this is often an insurmountable barrier.

The goal of this tutorial is to describe and illustrate a flexible Monte Carlo approach to power estimation that accommodates a broad class of multilevel regression models with continuous outcomes. Our approach adds to the literature in three important ways. First, it allows for any number of within-cluster effects, between-cluster effects, covariate effects at either level, cross-level interactions, and random coefficients. Moreover, we do not assume orthogonal effects, and predictors can correlate at either level. Second, our approach accommodates models with any number of interaction effects, and it does so with exact expressions for the variances and covariances of product random variables. Finally, we describe a strategy for deriving hypothetical population parameters that do not require pilot or comparable data. Instead, our approach uses intuitive variance-explained effect size expressions (Rights &

Sterba, 2019) to reverse-engineer solutions for the regression coefficients and variance components. We describe an R package `mlmpower` (Keller, 2023) that computes these solutions and automates the process of generating artificial data sets and summarizing the simulation results.¹

The structure of the article is as follows. First, we describe the two-level regression model and establish recurring notation. Second, we show how the multilevel model partitions the outcome variable's variation into within- and between-cluster components, each with distinct subparts. Third, we review the Rights and Sterba (2019) R^2 statistics that leverage the variance decompositions from the previous section. Fourth, we use the effect size expressions to derive solutions for the multilevel model's fixed effects and variance components. Fifth, we provide practical recommendations and guidelines for specifying the various inputs required by our solutions. Sixth, we demonstrate how to apply our approach to cross-sectional and longitudinal designs. Finally, we discuss some possible extensions of our method and suggest avenues for future research.

Multilevel Model and Notation

We focus on a two-level regression with a continuous outcome because this model enjoys widespread use in published applications. We use the following model as an exemplar

$$\begin{aligned}
 Y_{ij} = & (\beta_0 + b_{0j}) + (\beta_1 + b_{1j})(X_{1ij} - \bar{X}_{1j}) + \beta_2(X_{2ij} - \bar{X}_2) \\
 & + \beta_3(Z_{1j} - \bar{Z}_1) + \beta_4(Z_{2j} - \bar{Z}_2) \\
 & + \beta_5(X_{1ij} - \bar{X}_{1j})(Z_{1j} - \bar{Z}_1) + \varepsilon_{ij} \\
 \mathbf{b}_j \sim & N(0, \mathbf{\Sigma}_b) \quad \mathbf{\Sigma}_b = \begin{pmatrix} \sigma_{b_0}^2 & \\ \sigma_{b_1, b_0} & \sigma_{b_1}^2 \end{pmatrix} \quad \varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2),
 \end{aligned} \tag{1}$$

where Y_{ij} is the outcome score for observation i in cluster j , X_{1ij} is a focal within-cluster predictor, $\bar{X}_{1j}$ is the variable's Level 2 cluster mean, X_{2ij} is a grand mean-centered Level 1 covariate, Z_{1j} is a Level 2 moderator score for cluster j , and Z_{2j} is a Level 2 covariate. We do not include X_1 's cluster means in the model, but fundamentally the $\bar{X}_{1j}$ term variable is no different from "pure" Level 2 predictors. Turning to the residual terms, b_{0j} and b_{1j} are between-cluster random effects that capture residual variation in the cluster-specific intercepts and slopes, and ε_{ij} is a within-cluster residual. Going forward, we restrict random coefficients to focal predictors with no between-cluster variation (e.g., group mean-centered Level 1 predictors, or time score variables in a growth model). By assumption, the between-cluster random effects follow a multivariate normal distribution with a covariance matrix $\mathbf{\Sigma}_b$, and within-cluster residuals are normal with constant variance σ_ε^2 .

Our centering choices at Level 1 are deliberate and important. A brief review of group and grand mean centering establishes some key ideas for later in the article. To begin, grand mean centering continuous Level 2 predictors like Z_1 and Z_2 is usually a cosmetic operation. Importantly, we require centering at Level 2 because doing so simplifies the distributions of product random variables in a way that isolates their contribution to the explained variance. Consequently, we are able to define a unique effect size for the interaction effects.

Turning to the Level 1 predictors, centering X_1 at its Level 2 cluster means yields within-cluster deviation scores that are orthogonal to all sources of Level 2 variation (Enders & Tofighi, 2007). This

form of centering is often recommended for Level 1 random coefficient predictors because it isolates the within-cluster slope and avoids confounded variance components (Hamaker & Muthén, 2020). Importantly, group mean-centered predictors have an intraclass correlation equal to 0 because centering removes between-cluster variation (i.e., all group means equal 0 after centering). The time scores in a multilevel growth model are a second type of Level 1 predictor with no between-cluster variation. Time score predictors usually have nonzero means, but their properties are otherwise the same as group mean-centered regressors.

In contrast, grand mean centering defines X_2 as a composite Level 1 predictor that contains within- and between-cluster variation, as follows.

$$X_{2ij} - \bar{X}_2 = (X_{2ij} - \bar{X}_{2j}) + (\bar{X}_{2j} - \bar{X}_2) = X_{2ij}^w + X_{2j}^b. \tag{2}$$

Importantly, the intraclass correlation of a grand mean-centered variable can be nonzero, and variables like X_2 correlate with pure Level 2 predictors via their Level 2 group means. Finally, both the within- and between-cluster parts of the predictor can influence the outcome.

Although it need not be true in practice, we assume that the within- and between-cluster parts of grand mean-centered variables share a common slope coefficient, in which case Equation 1 can be rewritten as follows.

$$\begin{aligned}
 Y_{ij} = & (\beta_0 + b_{0j}) + (\beta_1 + b_{1j})X_{1ij}^w + \beta_2(X_{2ij}^w + X_{2j}^b) \\
 & + \beta_3Z_{1j}^b + \beta_4Z_{2j}^b + \beta_5X_{1ij}^wZ_{1j}^b + \varepsilon_{ij} \\
 = & (\beta_0 + b_{0j}) + (\beta_1 + b_{1j})X_{1ij}^w + \beta_2X_{2ij}^w \\
 & + \beta_2X_{2j}^b + \beta_3Z_{1j}^b + \beta_4Z_{2j}^b + \beta_5X_{1ij}^wZ_{1j}^b + \varepsilon_{ij}.
 \end{aligned} \tag{3}$$

Following Equation 2, the w and b superscripts reference within- and between-cluster deviation scores, respectively (i.e., group mean- and grand mean-centered variables, respectively). Importantly, notice that β_2 appears twice, indicating that X_2 's within- and between-cluster regression slopes are identical. When a model includes Level 1 predictors with nonzero intraclass correlations, our data-generating process explicitly models the within- and between-cluster components following Equation 3, as doing so allows for cross-level correlations (a realistic feature that previous studies have largely ignored).

Model-Predicted Outcome Variance

As explained previously, our approach derives population parameters from R^2 effect size expressions like those from Rights and Sterba (2019). We briefly review sources of explained and unexplained variance from a multilevel model, as this sets the stage for the effect size expressions in the next section. Additional details are available elsewhere in the literature (Rights & Sterba, 2019, Equation 10; T. A. B. Snijders & Bosker, 2012, Equation 7.9).

Although typical multilevel modeling applications do not invoke distributional assumptions for regressors, expressions for explained

¹ `mlmpower` is available at <https://cran.r-project.org/web/packages/mlmpower/index.html>.

variation require the variances and covariances of these variables. Our Monte Carlo simulation procedure explicitly invokes distributional assumptions for the predictors. For simplicity, we assume that continuous predictors are normally distributed, although the procedure also accommodates binary regressors at Level 2. Also for simplicity, we center all predictors, but Level 1 predictors can have nonzero means if desired. For example, the time scores from a multilevel growth model often have a nonzero mean (e.g., when the first measurement is coded 0).

The within-cluster parts of all Level 1 predictors have a variance–covariance matrix Φ_w , and the between-cluster parts of Level 1 predictors and pure Level 2 predictors have a covariance matrix Φ_b . The covariance matrices for the illustrative model from Equation 3 are shown below.

$$\Phi_w = \begin{pmatrix} \sigma_{X_1^w}^2 & \\ \sigma_{X_2^w, X_1^w} & \sigma_{X_2^w}^2 \end{pmatrix} \quad \Phi_b = \begin{pmatrix} \sigma_{X_2^b}^2 & & \\ \sigma_{Z_1^b, X_2^b} & \sigma_{Z_1^b}^2 & \\ \sigma_{Z_2^b, X_2^b} & \sigma_{Z_2^b, Z_1^b} & \sigma_{Z_2^b}^2 \end{pmatrix}. \quad (4)$$

Cross-level product terms have a distinct within-cluster covariance matrix Φ_p . Product random variables are unique because their expected values, variances, and covariances with other variables are fully determined by the lower-order moments of their components (Bohnstedt & Goldberger, 1969). For example, the variance of the cross-level product term in Equation 3 is a deterministic function of $\sigma_{X_1^w}^2$ and $\sigma_{Z_1^b}^2$, and its covariance with all other predictors equals 0 (this desirable feature follows from defining Level 2 predictors as normally distributed deviation scores). Although Φ_p is a scalar in this example, the covariance between any pair of cross-level product variables is also fully determined by the elements in Φ_w and Φ_b . Interested readers can find analytic details about product random variables in the online supplemental materials.

Having defined the necessary covariance matrices, we can now show how a multilevel model partitions variation in an outcome.

$$\text{var}(Y_{ij}) = \{\beta_w' \Phi_w \beta_w + \beta_p' \Phi_p \beta_p + \text{tr}(\Sigma_{b(rc)} \Phi_{w(rc)}) + \sigma_e^2\} + [\beta_b' \Phi_b \beta_b + \sigma_{b_0}^2]. \quad (5)$$

The expression separates regression slopes into three coefficient vectors: β_w and β_b contain the within- and between-cluster regression slopes, respectively, and β_p contains the cross-level interaction effects. In the running example, $\beta_w = (\beta_1, \beta_2)$, $\beta_b = (\beta_2, \beta_3, \beta_4)$, and $\beta_p = \beta_5$. Focusing on the term with the matrix trace, $\Sigma_{b(rc)}$ is the submatrix of the random effect covariance matrix Σ_b corresponding to the random slopes, and $\Phi_{w(rc)}$ is the submatrix of random predictors from Φ_w . Equation 5 is the denominator of the effect size expressions in the next section.

R^2 Effect Sizes

The curly braces in Equation 5 enclose four terms that define within-cluster variation: the first two terms reflect variance explained by the fixed effects of the Level 1 predictors and cross-level interactions, the third term is variance due to Level 1 predictors via their random slopes, and the fourth term is the within-cluster residual variance. Borrowing from Rights and Sterba (2019), the following equations express within-cluster variation due to predictors and random slopes as proportion variance

explained effect sizes.

$$R_w^2 = \frac{\beta_w' \Phi_w \beta_w}{\text{var}(Y_{ij})}. \quad (6)$$

$$R_p^2 = \frac{\beta_p' \Phi_p \beta_p}{\text{var}(Y_{ij})}. \quad (7)$$

$$R_{rc}^2 = \frac{\text{tr}(\Sigma_{b(rc)} \Phi_{w(rc)})}{\text{var}(Y_{ij})}. \quad (8)$$

The within-cluster residual variation can also be expressed as a proportion, but that conversion is not necessary here. To reiterate, centering Level 2 predictors simplifies the distributions of product random variables in a way that isolates their contribution to the explained variance, leading to the R_p^2 effect size in Equation 7.

The square brackets in Equation 5 contain two terms that determine between-cluster variation: the first term is variance explained by the Level 2 fixed effects, and the second term is the residual variation of the random intercepts. Rights and Sterba (2019) define a variance ratio for the random intercept variance, but our approach requires only the effect size for the fixed effects. When the Level 1 predictors have no between-cluster variation (e.g., group mean-centered predictors or time scores), the proportion explained variance attributable to Level 2 predictors is as follows.

$$R_b^2 = \frac{\beta_b' \Phi_b \beta_b}{\text{var}(Y_{ij})}. \quad (9)$$

In applications where Level 1 predictors have nonzero intraclass correlations, part of R_b^2 is inherited from the Level 1 predictors via their cluster means. Such is the case in the illustrative model, where β_2 appears in both β_w and β_b (see Equation 3). In this case, we define R_b^2 as the net variation explained by pure Level 2 predictors like Z_1^b and Z_2^b beyond that explained by the Level 2 cluster means. The effect size is

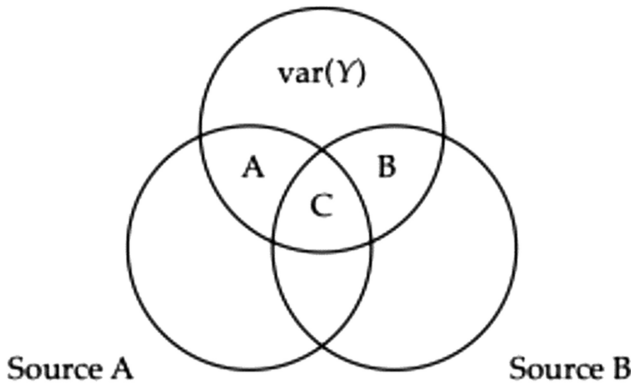
$$R_b^2 = \frac{\beta_b' \Phi_b \beta_b - \beta_{0b}' \Phi_b \beta_{0b}}{\text{var}(Y_{ij})}, \quad (10)$$

where β_{0b} is a coefficient vector from a nested model that fixes the cluster mean slopes to their within-cluster values and sets all other entries to 0. Looking at the Venn diagram in Figure 1, area A represents explained variation inherited from Level 1 predictors via their group means, and area B denotes unique variation attributable to pure Level 2 predictors. The R_b^2 effect size corresponds to area B.

Solving for the Multilevel Model Parameters

Conjuring reasonable population parameter values is a vital part of any power analysis. Absent pilot data, this step is virtually intractable because multilevel models have many moving parts and multiple sources of explained variance. The major contribution of our article is to simplify this process by casting the population parameters as functions of the R^2 effect size expressions from the previous section. The Rights and Sterba (2019) equations have covariance matrices and multilevel model parameters as their inputs, and R^2 values are the outputs. In our approach, covariance matrices and R^2 values are the inputs, and multilevel model parameters are the outputs. The resulting parameters define the population model for the power simulations.

Figure 1
Venn Diagram Depicting Sources of Explained Variation



Note. The sum of areas A, B, and C represent the explained variation in the outcome. Areas A and B represent the unique contribution of two predictors to the total effect, and area C is shared covariation between the predictors.

Solving for the multilevel model parameters requires the following inputs beyond the R^2 values: the intraclass correlations of all Level 1 variables, means and variances of all variables, within-cluster correlations among the Level 1 predictors, between-cluster correlations among the Level 2 predictors, and correlations among the random effects. Table 1 summarizes the inputs, catalogs our software's default settings, and provides recommendations for specifying values. At first glance, the number of inputs appears to be a daunting barrier that puts our approach beyond the grasp of practicing researchers. In fact, the effect size values and intraclass correlations are the most important parts; means and variances are completely arbitrary, and the various correlations tend not matter very much. Fortunately, published literature reviews often inform our a priori expectations about the important quantities.

The analytic work to derive the multilevel model parameters is not the focus of this tutorial article, so we summarize the solutions in the

Appendix and point interested readers to the online supplemental materials for a thorough description of the derivations. In the simplest case where the target model includes a single predictor at each level (and possibly an interaction term), the solutions are very simple and require little input from the researcher. With multiple predictors at each level, the R^2 values in the previous section are driven by multivariate inputs that introduce indeterminacies.

We obtain unique solutions for the model parameters by invoking weights that define the relative contribution of each predictor to a given source of explained variance. To illustrate, consider the Level 1 predictors from the model in Equation 1. Returning to the Venn diagram in Figure 1, suppose that areas A and B represent the unique contribution of X_1^w and X_2^w , respectively, to the within-cluster effect size R_w^2 , and area C is shared covariation. Assigning proportional weights to the pair of predictors imposes the constraints necessary to obtain unique solutions for β_1 and β_2 . For example, specifying $\omega_w = (0.50, 0.50)$ as a weight vector implies that both predictors contribute equally to explained variation, whereas setting $\omega_w = (0.75, 0.25)$ means that X_1^w 's unique contribution is about three times larger than X_2^w 's (i.e., area A is about three times larger than area B).

Analogous weights are required for the other effect sizes: ω_p is the weight vector for cross-level product terms that contribute to R_p^2 , ω_b weights the contribution of the pure Level 2 predictors to R_b^2 , and ω_{rc} weights the relative contributions of the random coefficient predictors to R_{rc}^2 . Returning to the model from Equation 3, $\omega_p = 1$ because there is a single cross-level product term, and $\omega_{rc} = 1$ because a single Level 1 predictor contributes to the random slope variation. As noted previously, ω_w contains a pair of weights that define the relative contributions of X_1^w and X_2^w , and ω_b similarly contains proportions that weight the unique contributions of Z_1^b and Z_2^b . The next section provides guidance about specifying these weights.

Selecting Inputs and Specifying a Model

Table 1 summarizes the inputs for our Monte Carlo power procedure. Although the number of inputs appears daunting, several are

Table 1
mlmpower Package Inputs and Recommendations

Input	Default	Recommendation
Within-cluster predictor R^2 (i.e., R_w^2)	None	Pilot data, previous studies, off-the-shelf values with sensitivity checks.
Cross-level product R^2 (i.e., R_p^2)	None	Pilot data, previous studies, off-the-shelf values with sensitivity checks.
Random coefficient R^2 (i.e., R_{rc}^2)	None	Pilot data, Table 2 values with sensitivity checks
Between-cluster predictor R^2 (i.e., R_b^2)	None	Pilot data, previous studies, off-the-shelf values with sensitivity checks.
Outcome variable ICC	None	Pilot data, previous studies, off-the-shelf ICCs with sensitivity checks.
Level-1 predictor ICC (group mean centered)	None	Must be set to 0.
Level-1 predictor ICC (grand mean centered)	None	Pilot data, previous studies, off-the-shelf ICCs with sensitivity checks.
Outcome mean	0	Arbitrary.
Level-1 predictor means	0	Arbitrary. Time score predictors may require a nonzero mean.
Level-2 predictor means	0	Fixed quantity. Must be 0.
Level-2 binary predictor group proportions	0.50/0.50	Pilot data, previous studies, expert judgment.
Outcome variance	1	Arbitrary.
Level-1 predictor variances	1	Arbitrary. Time score predictors may require a different variance.
Level-2 predictor variances	1	Arbitrary.
Within-cluster predictor correlations	0.10–0.30	Somewhat arbitrary. Tends not to be very impactful.
Between-cluster predictor correlations	0.10–0.30	Somewhat arbitrary. Tends not to be very impactful.
Random effect correlations	0.30	Somewhat arbitrary. Tends not to be very impactful.

Note. ICC = intraclass correlation.

either completely arbitrary or relatively unimportant. For less important quantities, the R package `mlmpower` (Keller, 2023) further simplifies model specification by invoking reasonable default parameter values, while retaining flexibility for tailoring power simulations. This section provides practical recommendations for selecting the necessary inputs, and it demonstrates the corresponding R specification and package defaults. Subsequent sections provide detailed illustrations for cross-sectional and longitudinal applications. As noted previously, Table 2 in Murayama et al. (2022) describes the inputs and assumptions for competing approaches in the literature

R^2 Effect Sizes

The R^2 effect sizes described earlier in the article are vital inputs to the computer simulation. Accordingly, the `mlmpower` package does not invoke default assumptions about these important quantities. Locating values from published multilevel articles probably is not a viable option because popular commercial software packages generally do not report R^2 effect sizes. However, a strength of the Monte Carlo approach is that its inputs need not be exact—approximate R^2 values can even be obtained from single-level studies with similar variables. To account for the inherent uncertainty about these inputs, one would simply conduct computer simulations across a range of realistic effect size values. We demonstrate this strategy later in the article.

Absent pilot data or a comparable data set with a similar structure and effects, researchers can use Cohen's (1988) f^2 effect size benchmarks for the within- and between-cluster regression coefficients. The relation between f^2 and R^2 is shown below.

$$f^2 = \frac{R^2}{1 - R^2} \quad R^2 = \frac{f^2}{1 + f^2}. \quad (11)$$

Cohen (1988) suggested that small effects range between $f^2 \geq 0.02$ and 0.15, medium effects range from $f^2 \geq 0.15$ to 0.35, and large effects align with $f^2 \geq 0.35$. On the variance explained metric, these thresholds correspond to $R^2 \geq 0.019$, 0.13, and 0.26.

These off-the-shelf effect size guidelines come with important caveats. First, the generalizability of Cohen's benchmarks may be questionable. Empirical studies suggest his guidelines may bear little resemblance to the distributions of effects observed in practice, which are often much smaller, on average (Bosco et al., 2015; Funder & Ozer, 2019; Gignac & Szodorai, 2016; Kraft, 2020). Second, even if they do apply to additive effects, applying off-the-shelf benchmarks to interactions is almost certainly inappropriate, as literature reviews suggest that moderation effects tend to explain rather small amounts of variance. For example, based on a review of 261 analyses from applied psychology and management journals, Aguinis et al. (2005) reported that the mean effect size was approximately $f^2 = 0.009$ ($R^2 \approx 0.0089$). Chaplin's (1991) review of published personality research similarly concluded that interaction effects typically explain about 1% of the variance. Until further evidence accumulates suggesting otherwise, limiting R_p^2 to values between 0.01 and 0.05 might be prudent.

Assigning an R^2 to within-cluster variation explained by the random coefficients is arguably the most difficult quantity to specify because few software packages report this relatively new effect size.² To provide researchers with practical guidelines, we attempted to locate the source data for random coefficient analysis examples in

several multilevel and longitudinal modeling textbooks. Some books had multiple examples, and some illustrations featured more than one random slope predictor. For each example, we group mean centered the random predictors and entered them individually to obtain unconditional estimates of R_{rc}^2 .

Table 2 summarizes the effect size estimates from each source text. Across all sources, the effect sizes ranged from 0.001 to 0.094, with an average of 0.038. It is noteworthy that the average effect size differed by data collection design; the mean R^2 value from the longitudinal applications was approximately 0.046 versus 0.024 for the cross-sectional applications. While it is tempting to conclude that trajectory differences are a more salient determinant of variation, the predictors from the cross-sectional examples were likely chosen for didactic rather than substantive reasons. Thus, it is unclear whether the tabled estimates are representative of published applications. We are somewhat more confident about the repeated measures estimates because those applications uniformly featured Level 1 time scores as the random predictor. Anecdotally, our experience with real data sets suggests the values in Table 2 are realistic and common, although we have occasionally observed much larger values (e.g., values of 15% or larger in a longitudinal clinical trial data). We recommend investigating multiple configurations of effect size values in order to account for uncertainty about these important quantities.

Intraclass Correlations

Like the R^2 effect size values, intraclass correlations are vital determinants of power that require careful consideration. Accordingly, the `mlmpower` package does not invoke default assumptions about intraclass correlations. Fortunately, published literature may inform our a priori expectations about these quantities. For example, a great deal of information exists about clustering effects in educational studies with students nested in schools. Several authoritative studies suggest that intraclass correlations for achievement outcomes routinely range between 0.10 and 0.25 (Hedges & Hedberg, 2007, 2013; Sellström & Bremberg, 2006; Spybrook et al., 2011; Stockford, 2009). Additionally, the Variance Almanac of Academic Achievement website³ is a useful resource that catalogs variance components and intraclass correlations from national achievement data sets spanning kindergarten through the 12th grade.

Guidelines for many other research contexts are also available in the literature. Several such examples come from the prevention and public health context: Murray and Short (1995) reported community-level intraclass correlations near 0.02 for a number of alcohol-related dependent variables; Pals et al. (2009) cataloged neighborhood-level intraclass correlations for various measures of sexual behavior between 0.02 and 0.07; Glassman et al. (2015) reported school-level intraclass correlations between 0 and 0.15 for outcomes in a prevention programming targeting sexually transmitted infections and pregnancy; Katzmarzyk et al. (2022) described typical clinic-level intraclass correlations between 0.02 and 0.05 for outcomes in a weight loss intervention program; and Masood and Reidpath (2016) reported

² Rights and Sterba's effect sizes are currently available from the R package `r2mlm` (Shaw et al., 2022) and the `Blimp` application (Keller & Enders, 2023).

³ <https://stemcenter.norc.org/reese/variance-almanac-academic-achievement.html>.

Table 2
Random Slope Effect Size Estimates From Selected Textbooks

Source	Type	R^2_{rc} estimates
Bolger and Laurenceau (2013)	Repeated measures	0.022, 0.057, 0.079, 0.080, 0.087
Finch et al. (2014)	Repeated measures	0.001, 0.019
Goldstein (2011)	Cross-sectional	0.046
Goldstein (2011)	Repeated measures	0.046
Grimm et al. (2016)	Repeated measures	0.009
Heck et al. (2022)	Cross-sectional	0.006
Heck et al. (2022)	Repeated measures	0.033
Hedeker and Gibbons (2006)	Repeated measures	0.066
S. W. Raudenbush and Bryk (2002)	Cross-sectional	0.007, 0.01
S. Raudenbush et al. (2019)	Cross-sectional	0.007, 0.014
S. Raudenbush et al. (2019)	Repeated measures	0.015, 0.064
Shiverdecker and LeBreton (2019)	Cross-sectional	0.091
Singer and Willett (2003)	Repeated measures	0.094
T. A. B. Snijders and Bosker (2012)	Cross-sectional	0.01
T. A. B. Snijders and Bosker (2012)	Repeated measures	0.018

Note. A cross-sectional example from Kreft and de Leeuw (1998) produced an effect size of 0.345. We omit this value from the table because it such a large aberration from typical. The authors note that they based the analysis on a very small subset of the NELS88 data set with only 10 Level 2 clusters.

country-level intraclass correlations between 0.04 and 0.19 for body mass index, physical activity, and diet-related outcomes.

Literature reviews from other disciplines provide similar guidelines. In clinical psychology, Kivlighan et al. (2020) reported an average intraclass correlation of 0.06 from a meta-analysis of group therapy studies with individuals nested in groups. Baldwin et al. (2011) conducted a similar comprehensive review of psychotherapy studies with patients nested in therapists, and these authors recommended using intraclass correlation values of 0.01–0.05 for power analyses.

Finally, repeated measures data sets with observations nested within persons tend to yield much larger intraclass correlations; Bolger and Laurenceau (2013) suggest that values between 0.20 and 0.40 are common in daily diary studies, and longitudinal studies may have even larger intraclass correlations in the range of 0.40–0.50 (Arend & Schäfer, 2019; Singer & Willett, 2003). These are just a few examples that highlight the breadth of information available in the literature. Again, we recommend investigating a range of intraclass correlation values in order to account for uncertainty in this important quantity. The `mlmpower` package facilitates these sensitivity checks by offering researchers the option to conduct simulations across a range of intraclass correlations.

Weighting Variables

With a single predictor at each level and one cross-level interaction—a common scenario in previous methodological studies—there is no need to allocate each variable’s contribution to a particular source of variation because all weights equal 1 (i.e., each effect size is driven by a single variable or product). However, realistic models with multiple predictors at each level require multivariate inputs that introduce indeterminacies. As explained previously, we obtain unique solutions for the model parameters by invoking weights that define the relative contribution of each predictor to a given source of explained variance. By default, the `mlmpower` package assigns equal weights within each group of variables. We implemented this setting to simplify the specification of models where each source of

variation is due to a single predictor (the predominant focus of previous power studies). We generally do not recommend equal weighting, although it could be reasonable for some scenarios (e.g., hypotheses involving an omnibus test of all predictors).

Our experiences consulting with substantive researchers suggest that prototypical regression models often feature a small number of focal predictors and an additional set of covariates or control variables. Variables in the former category are central to one’s research question, while those in the latter group are ancillary and would not figure prominently in a Results section. When thinking about weights, we find it useful to assign variables to these two buckets because it is usually much easier to make educated guesses about a binary allocation for a given R^2 value. Moreover, we usually start with the covariate set and then move to the focal predictors.

To illustrate the weighting process, consider the fixed effect R^2 values (R^2_w and R^2_b). One common scenario is a covariate set comprised of background variables and sociodemographic characteristics. These variables often have relatively weak associations with an outcome and do not explain a large pot of variation. Thus, it might be reasonable to assign 20% of the within- or between-cluster R^2 value to the covariate set and the remaining 80% to the focal predictors. Such an allocation would require covariate weights that sum to 0.20 and focal predictor weights that sum to 0.80. In other situations, the covariate set might include a pretest measure of the outcome, in which case the allocation to this group could be much larger, with covariate weights summing to 0.50 or higher. After making the initial binary allocations, we would simply distribute the covariate allocation equally across the member variables. For example, dividing 0.20 across a set of four background or control variables would result in variable-specific weights of 0.05. Having specified the covariate weights, assigning values to the focal predictors is usually simpler because most models have only one or two crucial regressors at each level. Absent specific knowledge about the relative magnitudes of their correlations with the outcome, it might again be reasonable to distribute the focal weight across those predictors.

Next, consider the effect size values for the cross-level interaction terms and random coefficients (R^2_p and R^2_{rc}). We suspect that many

(if not most) applications feature a single interaction and random coefficient, in which case the weights simply equal 1. In models with two or more interaction terms, equally weighting the product variables may be a reasonable default strategy. Alternatively, downweighting an interaction would yield conservative power estimates, and vice versa. We would generally apply the same strategy to models with two or more random coefficient predictors. An exception to this recommendation occurs with longitudinal growth curve models, where heterogeneity in the change rates is often a salient feature that could warrant a larger allocation. Again, the strength of computer simulation is that researchers can simply investigate multiple configurations of the key weights in order to account for uncertainty about these quantities.

Means and Standard Deviations

The means and total standard deviations of all variables are completely arbitrary, and specifying these quantities is optional. By default, the `mlmpower` package fixes the means and standard deviations of all variables to 0 and 1, respectively. Exceptions to this rule occur when using a Level 1 time score predictor or Level 2 binary predictor, in which case the software automatically determines the population means and standard deviations from the time scores and category proportions, respectively. As noted previously, Level 2 variables must have population means equal to 0 because this ensures that the cross-level product variables are orthogonal to all other predictors (see the online supplemental materials). Although there is generally no reason to change the default values, the package allows researchers to do so.

Correlations

The final inputs in Table 1 are three sets of correlations: within-cluster correlations among Level 1 predictors, between-cluster correlations among Level 2 predictors, and random effect correlations. Specifying these quantities is optional. By default, the `mlmpower` package randomly samples all correlations from a uniform distribution between 0.10 and 0.30, such that the resulting power estimates average over a distribution of possible associations. This default range spans Cohen's (1988) small to medium effect size benchmarks, and it also brackets common correlations from published research (Bosco et al., 2015; Funder & Ozer, 2019; Gignac & Szodorai, 2016), which tend to be in Cohen's small range. Although published correlations are likely larger due to publication bias (Bosco et al., 2015), that tendency would almost certainly lead to conservative power estimates because tests of partial regression slopes would benefit from lower rather than higher predictor correlations.

Based on the aforementioned literature reviews, we are generally comfortable using the default ranges for predictor correlations. Higher correlations may be warranted for Level 1 predictors in longitudinal applications, where time-varying predictors could have more salient associations with the time scores. Extrapolating these ranges to random effect correlations may be tenuous given the absence of solid guidelines from the literature. Some authors have argued that setting random effect correlations to 0 leads to conservative power estimates (Arend & Schäfer, 2019; Martin et al., 2011), and others suggest that fixed effect coefficients are not very sensitive to these correlations (T. A. Snijders & Bosker, 1993). Comparing power estimates from the default setting to those obtained by defining Σ_b as a diagonal matrix can provide a useful sensitivity check.

Illustration 1: Cross-Sectional Power Analysis

The first illustration demonstrates a power analysis for a cross-sectional application of the multilevel model from Equation 1. We sketch the main details for the example in this section and point interested readers to the online supplement materials for a detailed vignette. To provide a substantive context, consider a prototypical education example where students are nested in schools. As noted in the previous section, intraclass correlations for achievement-related outcomes often range between 0.10 and 0.25. To accommodate uncertainty about this important parameter, we investigated intraclass correlation values of 0.10 and 0.25.

Turning to the R^2 effect sizes, we adopted Cohen's medium effect size benchmark for the fixed effects ($f^2 = 0.15$, $R^2 = 0.13$), and we distributed this variation evenly at the within- and between-cluster level: $R_w^2 = 0.065$ and $R_b^2 = 0.065$. Note that explained variation at the between-cluster level will be slightly larger than 0.065 because this proportion reflects the net increment beyond variation explained by the Level 2 cluster means in X_2^b . As mentioned previously, R_b^2 cannot exceed the intraclass correlation, as this implies that the Level 2 predictors explain more than 100% of the available variation. Following conventional wisdom about interaction effects (Aguinis et al., 2005; Chaplin, 1991), we assigned $R_p^2 = 0.01$ to the cross-level product. Finally, we used the values in Table 2 to inform our decision about the random slope effect size, which we set to $R_{rc}^2 = 0.03$.

By default, `mlmpower` assigns equal weights to all quantities contributing to a particular source of variance. To assign nonuniform weights, we designated X_1 and Z_1 (the interacting variables) as focal predictors and X_2 and Z_2 as covariates. To mimic a situation where the covariates explain a small amount of variation, we assigned weights of 0.80 to the focal predictors and 0.20 to the covariates. As mentioned previously, a small covariate allocation could be appropriate for a set of background or sociodemographic characteristics that weakly predict the outcome. Researchers often need power estimates for individual partial regression slopes. Although the weights do not exactly carve R_w^2 and R_b^2 into additive components when predictors are correlated, adopting weak associations among the regressors allows us to infer that each focal predictor roughly accounts for 5% of the explained variation at each level (i.e., $0.80 \times R_w^2$ or $R_b^2 \approx 0.05$).

The following code block shows the `mlmpower` syntax for the example.

```
example1 <- (
  effect_size(
    icc          = c(.10, .25),
    within       = .065,
    between      = .065,
    product      = .01,
    random_slope = .03
  )
  + outcome('y', mean = 50, sd = 10)
  + within_predictor('x1', icc = 0, weight = .80)
  + within_predictor('x2', weight = .20)
  + between_predictor('z1', weight = .80)
  + between_predictor('z2', weight = .20)
  + product('x1', 'z1', weight = 1)
  + random_slope('x1', weight = 1)
)
```

The `mlmpower` package groups R^2 values and intraclass correlations into a single object called `effect_size()`. The five inputs are separated by commas, as shown above. The `icc` parameter assigns a global intraclass correlation to *all* Level 1 variables. The variable-specific attributes described below allow users to modify each variable's intraclass correlation, if desired. To facilitate sensitivity checks across a range of intraclass correlations, the `icc` parameter can also be a vector, as it is in this example. With vector input, the package performs a separate power simulation for each intraclass correlation. It is important to note that the between-cluster effect size given by the `between` argument cannot exceed the intraclass correlation, as such a specification implies that Level 2 predictors explain more than 100% of the between-cluster variance.

Next, variable-specific attributes are referenced by adding the following objects: `outcome()`, `within_predictor()`, `between_predictor()`, `product()`, and `random_slope()`. To begin, the `outcome()` object overrides the default setting by specifying a mean and standard deviation of 50 and 10 for the dependent variable, respectively. Assigning a larger standard deviation to the outcome is an aesthetic operation that avoids small, fractional slope coefficients.

As noted previously, the `effect_size()` object assigns a global intraclass correlation to all Level 1 variables, and the user can modify this setting on a variable-by-variable basis. Importantly, the `icc=0` argument in X_1 's `within_predictor()` object defines this regressor as a pure within-cluster deviation score (e.g., a group mean-centered predictor). In contrast, the global setting is adopted by omitting the `icc` argument from the `outcome()` object and X_2 's `within_predictor()` object.

Next, the weight arguments implement the aforementioned weighting scheme. Note that the interaction and random coefficient and interaction weights equal unity because a single term defines these sources of variation. Finally, with the exception of the outcome variable, which has a mean and standard deviation of 50 and 10, the script accepts the default settings for the means and variances of the predictors (0 and 1, respectively). As a reminder, predictor correlations are randomly sampled from a uniform range between 0.10 and 0.30.

It may be instructive to inspect the population parameters prior to running the simulation. Executing `summary(example1)` returns tabular summaries of the solutions. To illustrate, the values below result from setting the intraclass correlation to 0.25 and fixing all correlations to their mean value of 0.20.

$$\begin{aligned}
 Y_{ij} &= 50 + 2.36(X_{1ij}^w) + 0.68(X_{2ij}^w + X_{2j}^b) \\
 &\quad + 2.43(Z_{1j}^b) + 0.61(Z_{2j}^b) + 1.00(X_{1ij}^w)(Z_{1j}^b) + b_{0j} + b_{1j}X_{1ij}^w + \varepsilon_{ij} \\
 \Sigma_b &= \begin{pmatrix} 17.60 & \\ 1.45 & 3.00 \end{pmatrix} \quad \sigma_\varepsilon^2 = 64.5 \\
 \Phi_w &= \begin{pmatrix} 1.00 & \\ 0.17 & 0.75 \end{pmatrix} \quad \Phi_p = 1.00 \quad \Phi_b = \begin{pmatrix} 0.25 & & \\ 0.10 & 1.00 & \\ 0.10 & 0.20 & 1.00 \end{pmatrix}.
 \end{aligned} \tag{12}$$

Having specified the target model, you next use the `power_analysis()` function to conduct the simulations. The function requires four inputs: the model argument specifies the parameter value object (e.g., `example1`), the `replications` input specifies the number of artificial data sets, `n_between` is a vector of Level 2 sample

sizes, and `n_within` is a vector of Level 1 sample sizes (i.e., the number of observations per cluster). The code block below pairs four Level 2 sample sizes ($J = 30, 60, 90$, and 120) with three Level 1 sample sizes ($n_j = 10, 20$, or 30 observations per cluster), and it requests 2,000 artificial data sets for each combination.

```

powersim <-
  power_analysis(
    model = example1,
    replications = 2000,
    n_between = c(30,60,90,120),
    n_within = c(10,20,30)
  )

```

The data generation code behind the `power_analysis()` function performs the following steps: (a) sample Level 2 predictor scores for each cluster, (b) sample Level 1 predictor scores for each observation, (c) combine the within- and between-cluster parts of all covariates, (d) compute product terms by multiplying Level 1 and Level 2 predictors, (e) sample random intercepts and slopes for each cluster, (f) sample a within-cluster residual for each observation, and (g) compute the dependent variable as a weighted composite of the predictors and residual terms. The package uses `lme4` (Bates et al., 2021) for model fitting, and it defaults to an α level of .05 for all significance tests. Significance tests of random slope variation use a likelihood ratio test with a mixture chi-square reference distribution (i.e., a chi-bar distribution; T. A. B. Snijders & Bosker, 2012, p. 99), as implemented in the `varTestnlme` package (Baey & Kuhn, 2022). Executing `summary(powersim)` returns tabular summaries of the simulation results.

Table 3 shows the simulation-based power estimates for each parameter and sample size combination at intraclass correlation values of 0.10 and 0.25. The table highlights that power values vary considerably across model parameters. For example, the cross-level interaction effect was sensitive to both sample sizes, and power generally met or exceeded optimal 0.80 levels with $J = 60$ Level 2 units and $n_j \geq 20$ observations per cluster and nearly all conditions where $J \geq 90$. The intraclass correlation had a modest impact on power values, which were slightly higher in the 0.25 condition (e.g., differences of 3%–5% were typical). As a second example, the test of the Level 2 random slope variation was also impacted by both sample sizes, but the number of within-cluster observations was a stronger determinant.⁴ A larger intraclass correlation noticeably improved power. As a third example, power estimates for the Level 2 predictors decreased as the intraclass correlation increased. This finding might be counterintuitive in light of statements from the literature suggesting that power for Level 2 effects increases with the intraclass correlation (Scherbaum & Pesner, 2019). However, it is important to emphasize that our procedure uses unconditional intraclass correlations as inputs. Thus, increasing the intraclass correlation while

⁴ The null hypothesis model is a random intercept model where both the slope variance and the intercept–slope covariance equal 0. It is not possible to decouple the two parameters when using the likelihood ratio test because the absence of slope variation implies that the covariance must be 0.

keeping R_b^2 constant implies that the Level 2 predictors explain a smaller proportion of the available between-cluster variation. Collectively, the results highlight that one-size-fits-all rules are not very accurate, and they emphasize the importance of tailoring power estimates for a particular target model. For example, the oft-cited “30/30” rule of thumb (Kreft, 1996) produced a subpar power for the cross-level interaction but roughly adequate power for the random slope variation.

The table also conveys the impact of weighting choices. Comparing estimates from the Z_1^b and Z_2^b rows, power differences are entirely due to their differential weights of 0.80 and 0.20, respectively (recall that Z_1 was a focal predictor and Z_2 was a weak covariate). Assigning equal weights would have produced identical power estimates. Power differences between the Level 1 predictors are also driven by weighting, although not entirely. Given identical weights, X_2 's power would be somewhat higher because it explains both within- and between-cluster variation, whereas X_1 's effect is relegated to the within-cluster variability.

We previously claimed that predictor and random effect correlations tend not to matter very much. To illustrate this point, we performed a second set of simulations that used the `correlations()` object to fix all three sets of correlation values (within-cluster predictors, between-cluster predictors, and random effects) to 0. The `mlmpower` model object for the simulation is shown below.

```
example1 <- (
  effect_size(
    icc      = c(.10, .25),
    within   = .065,
    between  = .065,
    product  = .01,
    random_slope = .03
  )
  + outcome('y', mean = 50, sd = 10)
  + within_predictor('x1', icc = 0, weight = .80)
  + within_predictor('x2', weight = .20)
  + between_predictor('z1', weight = .80)
  + between_predictor('z2', weight = .20)
  + product('x1', 'z1', weight = 1)
  + random_slope('x1', weight = 1)
  + correlations(
    within = 0,
    between = 0,
    randeff = 0
  )
)
```

In the interest of space, we point readers to Table A2 in the online supplemental materials for a tabular summary of the power estimates. Across intraclass correlation and sample size combinations, the mean absolute difference between the two sets of power values was only 0.01 with a standard deviation of 0.02. The largest differences were associated with X_2 's slope, where eliminating correlations improved power by about 0.04, on average. This pattern makes sense, as X_2 has the most to gain by eliminating shared variation because it correlates with all other predictors.

Illustration 2: Growth Curve Power Analysis

The second illustration demonstrates a power analysis for a relatively complex longitudinal growth curve model with a pair of cross-level interactions and a binary Level 2 moderator. We again sketch the main details for the example and refer interested readers to the online supplement materials for a detailed vignette. The explanatory variables include a time score predictor with a random coefficient, a pair of time-varying covariates, a binary Level 2 moderator, a pair of Level 2 covariates, and a cross-level (group-by-time) interaction. The target model is as follows.

$$Y_{ij} = (\beta_0 + b_{0j}) + (\beta_1 + b_{1j})X_{1ij}^w + \beta_2(X_{2ij}^w + X_{2j}^b) + \beta_3(X_{3ij}^w + X_{3j}^b) + \beta_4Z_{1j}^b + \beta_5Z_{2j}^b + \beta_6Z_{3j}^b + \beta_7X_{1ij}^wZ_{1j}^b + \varepsilon_{ij} \quad (13)$$

For the purposes of weighting, we designated X_1 and Z_1 (the interacting variables) as focal predictors, X_2 and X_3 as Level 1 (time-varying) covariates, and Z_2 and Z_3 as Level 2 covariates. To mimic the scaling of a typical temporal index, we assume the major time increments are coded $X_1^w = (0, 1, 2, 3, 4)$.

A brief discussion of Z_1 (the binary moderator) is warranted before continuing. First, like other Level 2 variables, this variable's population mean must be 0 (i.e., Z_1 is centered in the population) in order to maintain the orthogonality of the cross-level interaction terms. Grand mean centering a Level 2 binary predictor creates an ANOVA-like contrast code, such that β_0 is the grand mean and β_4 is the predicted group mean difference when X_1^w (the time score predictor) equals 0. In this case, a code of 0 corresponds to the baseline assessment. Because β_4 is a conditional effect that represents the group mean difference at the first occasion, Z_1 's contribution to the between-cluster effect size depends on whether we view this regressor as a naturally occurring classification or an intervention assignment indicator. In the former case, we might expect a group mean difference at baseline, and the presence of such a difference would require a nonzero weight. In contrast, random assignment to conditions would eliminate a group mean difference at baseline, and Z_1 's weight would equal 0. Note that this conclusion changes if the time scores are coded differently. For illustration purposes, we assume Z_1 is an intervention assignment indicator.

As noted previously, intraclass correlations for repeated measures data are much higher, often exceeding 0.40. To accommodate uncertainty about this important parameter, we investigated intraclass correlation values of 0.40 and 0.60. Turning to the R^2 effect sizes, we assigned $R_w^2 = 0.13$ to the three within-cluster predictors (Cohen's medium effect size benchmark), and we assigned $R_b^2 = 0.065$ to the three Level 2 predictors (half the medium effect size benchmark). Note that R_b^2 cannot exceed the outcome's intraclass correlation, as this would imply that between-cluster predictors explain more than 100% of the available variance. We additionally set the cross-level (i.e., group-by-time) interaction effect size to $R_p^2 = 0.05$. In this world, nothing is certain but death, taxes, and slope heterogeneity in a growth curve analysis. Consequently, we set the effect size for the random coefficient variation to a somewhat higher value of $R_{rc}^2 = 0.10$.

To refresh, we designated X_1 and Z_1 (the interacting variables) as focal predictors, X_2 and X_3 as Level 1 (time-varying) covariates, and Z_2 and Z_3 as Level 2 covariates. To mimic a situation where the linear

change predominates the Level 1 model, we used weights of 0.50, 0.25, and 0.25 to allocate the within-cluster R^2 to X_1 , X_2 , and X_3 . At Level 2, we used weights equal to 0, 0.50, and 0.50 to allocate the between-cluster R^2 to Z_1 , Z_2 , and Z_3 . As noted previously, Z_1 's slope represents the group mean difference at baseline, and we are assuming that random assignment nullifies this effect. Finally, R_p^2 and R_{rc}^2 do not require weights because a single variable determines each source of variation.

To illustrate how to modify default correlation settings, we sampled within-cluster predictor correlations between the range of 0.20 and 0.40 under the assumption that the time scores could have stronger associations with other time-varying predictors. We similarly sampled the random effect correlations between the range of 0.30 and 0.50 to mimic a scenario where higher baseline scores (i.e., random intercepts) are associated with higher (more positive) growth rates. Finally, we adopted the default correlation range for the between-cluster predictors. The simulations from the previous cross-sectional example suggest that the correlations are somewhat arbitrary and would not have a material impact on power estimates.

The following code block shows the `mlmpower` model object for this example.

```
example2 <- (
  effect_size(
    icc      = c(.40,.60),
    within   = .13,
    between  = .065,
    product  = .03,
    random_slope = .10
  )
  + outcome('y', mean = 50, sd = 10)
  + within_time_predictor('x1', weight = .50, values = 0:4)
  + within_predictor('x2', weight = .25)
  + within_predictor('x3', weight = .25)
  + between_binary_predictor('z1', proportion = .50, weight = 0)
  + between_predictor('z2', weight = .50)
  + between_predictor('z3', weight = .50)
  + product('x1','z1', weight = 1)
  + random_slope('x1', weight = 1)
  + correlations(
    within = random(.20, .40),
    between = random(.10, .30),
    randeff = random(.30, .50)
  )
)
```

Three features are important to highlight. First, the `within_time_predictor()` object specifies X_1 as a temporal predictor with the fixed set of time scores described earlier. The `values = 0:4` argument specifies an integer sequence, but unequally spaced increments can also be specified using a vector as input (e.g., `values = c(0, 1, 3, 6)`). Second, we used the `between_binary_predictor()` object to define Z_1 (the moderator) as a dichotomous predictor, and we used the `proportion = 0.50` argument to induce a 50/50 split between the two groups. As explained previously, this variable's weight was set to 0 to mimic random assignment. Note that simulating binary variables requires a slight modification that attenuates between-cluster population correlations to account for categorization. The online supplemental materials describe the procedure. Finally, with the exception of the outcome,

the code block uses default values for all means and standard deviations (0 and 1, respectively). The means and standard deviations of the time scores and binary predictor are automatically determined by the user inputs.

Again, it may be instructive to inspect the population parameters prior to running the simulation. Executing `summary(example2)` returns tabular summaries of the multilevel model parameters. To illustrate, the values below result from setting the intraclass correlation to 0.60 and fixing correlations at their mean values.

$$\begin{aligned}
 Y_{ij} &= 46.60 + 1.70(X_{1ij}^w) + 1.90(X_{2ij}^w + X_{2j}^b) + 1.90(X_{3ij}^w + X_{3j}^b) + 0(Z_{1j}^b) \\
 &\quad + 1.74(Z_{2j}^b) + 1.74(Z_{3j}^b) + 2.45(X_{1ij}^w)(Z_{1j}^b) + b_{0j} + b_{1j}X_{1ij}^w + \varepsilon_{ij} \\
 \Sigma_b &= \begin{pmatrix} 15.90 & & \\ 3.57 & 5.00 & \\ & & \end{pmatrix} \quad \sigma_e^2 = 14 \\
 \Phi_w &= \begin{pmatrix} 2.00 & & \\ 0.27 & 0.40 & \\ 0.27 & 0.12 & 0.40 \end{pmatrix} \quad \Phi_p = 0.50 \\
 \Phi_b &= \begin{pmatrix} 0.60 & & & & \\ 0.18 & 0.60 & & & \\ 0.08 & 0.08 & 0.25 & & \\ 0.15 & 0.15 & 0.08 & 1.00 & \\ 0.15 & 0.15 & 0.08 & 2.00 & 1.00 \end{pmatrix}.
 \end{aligned} \tag{14}$$

Having specified the target model, you next use the `power_analysis()` function to conduct the simulations. We investigated power for six Level 2 sample size conditions ($J = 50, 60, 70, 80, 90$, and 100), each with five repeated measurements and 2,000 replications.

```
powersim <-
  power_analysis(
    model = example2,
    replications = 2000,
    n_between = c(50,60,70,80,90,100),
    n_within = 5
  )
```

Table 4 shows the power estimates for intraclass correlation values of 0.40 and 0.60. To begin, power estimates for the time score predictor (the X_1^w row) and the cross-level interaction were sufficient at every sample size. Consistent with the previous cross-sectional example, increasing the intraclass correlation improved the power to detect the interaction effect. Note that the very low values in the Z_1^b column are Type I error rates rather than power estimates because the true value of this coefficient was 0 (i.e., random assignment eliminated a baseline group mean difference). Owing to the relatively large effect size ($R_{rc}^2 = 0.10$), the power to detect random slope variation was uniformly 1.0.

We previously mentioned that the R^2 associated with random coefficients is a difficult quantity to specify because few software packages report this effect size. In applications where random

Table 3
Power Estimates From the Cross-Sectional Power Simulations

Sample sizes ICC	$J = 30$ $n_j = 10$		$J = 30$ $n_j = 20$		$J = 30$ $n_j = 30$		$J = 60$ $n_j = 10$		$J = 60$ $n_j = 20$		$J = 60$ $n_j = 30$	
	0.10	0.25	0.10	0.25	0.10	0.25	0.10	0.25	0.10	0.25	0.10	0.25
X_1^w Slope	0.92	0.96	0.99	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
X_2^{w+b} Slope	0.22	0.25	0.37	0.42	0.52	0.59	0.40	0.44	0.63	0.70	0.81	0.85
Z_1^b Slope	0.91	0.66	0.98	0.71	0.99	0.74	1.00	0.93	1.00	0.96	1.00	0.97
Z_2^b Slope	0.14	0.10	0.23	0.11	0.28	0.12	0.26	0.16	0.41	0.17	0.48	0.17
$X_1^w Z_1^b$ Slope	0.32	0.33	0.48	0.51	0.57	0.61	0.56	0.61	0.77	0.82	0.86	0.89
$\Sigma_{b(rc)}$	0.16	0.25	0.45	0.62	0.73	0.84	0.32	0.44	0.78	0.89	0.95	0.99

Sample sizes ICC	$J = 90$ $n_j = 10$		$J = 90$ $n_j = 20$		$J = 90$ $n_j = 30$		$J = 120$ $n_j = 10$		$J = 120$ $n_j = 20$		$J = 120$ $n_j = 30$	
	0.10	0.25	0.10	0.25	0.10	0.25	0.10	0.25	0.10	0.25	0.10	0.25
X_1^w Slope	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
X_2^{w+b} Slope	0.55	0.58	0.81	0.87	0.94	0.96	0.66	0.73	0.90	0.94	0.98	0.99
Z_1^b Slope	1.00	0.98	1.00	0.99	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Z_2^b Slope	0.37	0.21	0.55	0.24	0.62	0.23	0.47	0.24	0.66	0.29	0.75	0.31
$X_1^w Z_1^b$ Slope	0.73	0.80	0.93	0.95	0.97	0.97	0.86	0.91	0.98	0.99	0.99	0.99
$\Sigma_{b(rc)}$	0.46	0.64	0.91	0.98	0.99	1.00	0.58	0.75	0.97	1.00	1.00	1.00

Note. ICC = intraclass correlation, J = the number of Level 2 units, n_j = the number of Level 1 observations within each Level 2 cluster, and $\Sigma_{b(rc)}$ references the random slope variance and the covariance between the intercepts and slopes. Power estimates are the proportion of 2,000 data sets that produced a significant test statistic at $p < .05$.

slope variation is a key feature (e.g., models with cross-level interaction effects), we recommend sensitivity analyses that vary this parameter. The first simulation used an effect size that was somewhat larger than the typical value from Table 2. We therefore performed an additional simulation that specified $R_{rc}^2 = 0.05$. Table 5 shows the power estimates from simulations that set the intraclass correlation to 0.40. Decreasing the effect size had very little impact on the power to detect random slope variation, which was still nearly 1.0 in all cases. As one might expect, varying the random coefficient effect size influenced power estimates for the cross-level interaction (the term responsible for explaining random slope variation). In particular, notice that power estimates for the moderation effects decreased as the R_{rc}^2 increased. Increasing the proportion of residual random slope variation while keeping R_p^2 constant implies that the cross-level interactions explain a smaller amount of the total

available slope variation, the magnitude of which is determined by the sum of R_p^2 and R_{rc}^2 .

Illustration 3: Power for a Cluster-Randomized Design

The third vignette demonstrates a power analysis for a cluster-randomized design (S. W. Raudenbush, 1997) where Level 2 units are randomly assigned to one of two experimental conditions. To provide a substantive context, consider a prototypical education example where students are nested in schools, and schools are the unit of randomization. Intraclass correlations for achievement-related outcomes often range between 0.10 and 0.25 (Hedges & Hedberg, 2007, 2013; Sellström & Bremberg, 2006; Spybrook et al., 2011; Stockford, 2009). To accommodate uncertainty about this important parameter, the simulation investigates intraclass correlation values of 0.10 and 0.25.

Table 4
Power Estimates From the Longitudinal Power Simulations

Sample sizes ICC	$J = 50$		$J = 60$		$J = 70$		$J = 80$		$J = 90$		$J = 100$	
	0.40	0.60	0.40	0.60	0.40	0.60	0.40	0.60	0.40	0.60	0.40	0.60
X_1^w Slope	0.97	0.99	0.99	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
X_2^{w+b} Slope	0.84	0.99	0.91	1.00	0.94	1.00	0.97	1.00	0.98	1.00	0.99	1.00
X_3^{w+b} Slope	0.85	0.99	0.90	0.99	0.93	1.00	0.97	1.00	0.98	1.00	0.98	1.00
Z_1^b Slope	0.05	0.05	0.05	0.05	0.05	0.06	0.05	0.05	0.05	0.05	0.06	0.05
Z_2^b Slope	0.68	0.64	0.75	0.71	0.82	0.77	0.88	0.84	0.90	0.87	0.94	0.92
Z_2^b Slope	0.66	0.62	0.77	0.72	0.83	0.79	0.88	0.83	0.91	0.88	0.94	0.90
$X_1^w Z_1^b$ Slope	0.81	0.91	0.89	0.96	0.94	0.98	0.96	0.99	0.98	0.99	0.99	1.00
$\Sigma_{b(rc)}$	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Note. ICC = intraclass correlation, J = the number of Level 2 units, n_j = the number of Level 1 observations within each Level 2 cluster, and $\Sigma_{b(rc)}$ references the random slope variance and the covariance between the intercepts and slopes. Power estimates are the proportion of 2,000 data sets that produced a significant test statistic at $p < .05$.

Table 5*Power Estimates From the Longitudinal Power Simulations With ICC = 0.40 and Varying R_{rc}^2*

Sample sizes R_{rc}^2	$J = 50$		$J = 60$		$J = 70$		$J = 80$		$J = 90$		$J = 100$	
	0.05	0.10	0.05	0.10	0.05	0.10	0.05	0.10	0.05	0.10	0.05	0.10
X_1^w Slope	0.99	0.97	0.99	0.99	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
X_2^{w+b} Slope	0.79	0.84	0.85	0.91	0.91	0.94	0.93	0.97	0.96	0.98	0.98	0.99
X_3^{w+b} Slope	0.77	0.85	0.84	0.90	0.89	0.93	0.92	0.97	0.95	0.98	0.96	0.98
Z_1^b Slope	0.05	0.05	0.04	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.06
Z_2^b Slope	0.52	0.68	0.60	0.75	0.67	0.82	0.74	0.88	0.78	0.90	0.84	0.94
Z_3^b Slope	0.52	0.66	0.60	0.77	0.67	0.83	0.75	0.88	0.79	0.91	0.83	0.94
$X_1^w Z_1^b$ Slope	0.92	0.81	0.95	0.89	0.98	0.94	0.99	0.96	0.99	0.98	1.00	0.99
$\Sigma_{b(rc)}$	0.94	1.00	0.97	1.00	0.99	1.00	0.99	1.00	1.00	1.00	1.00	1.00

Note. R_{rc}^2 = proportion within-cluster variation explained due to random coefficients, J = the number of Level 2 units, n_j = the number of Level 1 observations within each Level 2 cluster, and $\Sigma_{b(rc)}$ references the random slope variance and the covariance between the intercepts and slopes. Power estimates are the proportion of 2,000 data sets that produced a significant test statistic at $p < .05$. ICC = intraclass correlation.

The multilevel model for the illustration is

$$Y_{ij} = \beta_0 + \beta_1(X_{1ij}^w + X_{1j}^b) + \beta_2(X_{2ij}^w + X_{2j}^b) + \beta_3(X_{3ij}^w + X_{3j}^b) + \beta_4(X_{4ij}^w + X_{4j}^b) + \beta_5 Z_{1j}^b + b_{0j} + \varepsilon_{ij}, \quad (15)$$

where X_1 to X_4 are grand mean-centered Level 1 covariates, and Z_1 is a binary intervention assignment indicator. Following established notation, the w and b superscripts reference within- and between-cluster deviation scores, respectively. The notation conveys that all Level 1 predictors contain within- and between-cluster variation. By default, all predictors are centered, including the binary dummy code. Grand mean centering a Level 2 binary predictor creates an ANOVA-like contrast code, such that β_0 is the grand mean and β_5 is the predicted mean difference, adjusted for the covariates. Turning to the residual terms, b_{0j} is a between-cluster random effect that captures residual variation in the cluster-specific intercepts, and ε_{ij} is a within-cluster residual. By assumption, both residuals are normal with constant variance.

The key inputs to the model object are the unconditional intraclass correlation and effect size values. This illustration investigates power for intraclass correlation values of 0.10 and 0.25. To mimic a scenario with a strong covariate set (e.g., one that includes a pretest measure of the outcome), the within-cluster effect size is set to $R_w^2 = 0.18$ (a value roughly midway between Cohen's medium and large benchmarks). Most of this variation was allocated to X_1 (the pretest) by assigning it a weight of 0.70, and the remaining three predictors had their within-cluster weights set to 0.10. We previously argued that the allocation of the weights among covariates is arbitrary because these predictors are not the focus. To demonstrate that point, we conducted a second simulation that weighted the four Level 1 covariates equally.

Turning to the Level 2 predictor, researchers often prefer Cohen's (1988) d effect size when working with binary explanatory variables. To illustrate, consider power for $d = 0.40$, the midway point between Cohen's small and medium benchmarks. Substituting this value into the conversion equation below returns $R_b^2 = 0.038$.

$$R^2 = \frac{d^2}{d^2 + 4}. \quad (16)$$

This effect size unambiguously aligns with the test of the group mean difference, adjusted for Level 1 covariates. Although it need not be the case, we assume random assignment leads to a 50/50 split in the binary classification variable.

The following code block shows the `mlmpower` model object for the example.

```
example3 <- (
  effect_size(
    icc          = c(.10, .25),
    within       = .18,
    between      = .038,
  )
  + outcome('y')
  + within_predictor('x1', weight = .70)
  + within_predictor('x2', weight = .10)
  + within_predictor('x3', weight = .10)
  + within_predictor('x4', weight = .10)
  + between_binary_predictor('z1', proportion = .50, weight = 1)
)
```

For brevity, we point readers to the detailed vignette for additional details about this simulation. Table A3 in the online supplemental materials provides power summaries for four Level 2 sample size values ($J = 30, 60, 90$, and 120) crossed with two Level 1 sample sizes ($n_j = 15$ or 30 observations per cluster). Table A4 in the online supplemental materials shows power summaries for the same scenario with equally weighted Level 1 covariates. As expected, weighting choices had no impact on the power to detect the Level 2 group mean difference (weighting did affect power to detect covariate effects, not surprisingly). This finding supports our earlier recommendation to distribute the covariate allocation equally among the member variables.

Discussion

Estimating power for multilevel models is complex because there are many moving parts, several sources of explained variation to consider, and unique sample sizes at Level 1 and Level 2. Monte Carlo computer simulation is a flexible power estimation tool that has received considerable attention in the literature. However,

much of the work to date has focused on very simple models with one predictor at each level and one cross-level interaction effect, and approaches that do not share this limitation require the user to input a typically-large set of presumed population parameters.

The goal of this tutorial was to describe and illustrate a flexible Monte Carlo approach to power estimation that accommodates a broad class of multilevel regression models with continuous outcomes. Our approach makes three important contributions. First, it allows for any number of within-cluster effects, between-cluster effects, covariate effects at either level, cross-level interactions, and random coefficients. Moreover, we do not assume orthogonal effects, and predictors can correlate at either level. Second, our approach accommodates models with multiple interaction effects, and it does so with exact expressions for the variances and covariances of product random variables. Finally, we described a strategy for deriving hypothetical population parameters that does not require pilot or comparable data. Instead, our approach uses intuitive variance-explained effect size expressions (Rights & Sterba, 2019) to reverse-engineer solutions for the regression coefficients and variance components. We provide an R package `mlmpower` that computes these solutions and automates the process of generating artificial data sets and summarizing simulation results.

While our approach substantially broadens the scope of possibilities with Monte Carlo power simulations, a number of avenues for future research remain. First, our approach could be extended to include binary outcomes by adopting a probit regression formulation that views categorical responses as arising from a normally distributed latent response variable. Although it requires slightly different constraints (e.g., the within-cluster residual variance of the outcome is fixed rather than its total variance), this extension fits neatly into our framework. Second, we limited our focus to balanced designs with a constant number of Level 1 observations within each Level 2 cluster, but minor modifications could induce unequal group sizes and other types of missing data. Finally, our method readily extends to three-level regression models, which leverage analogous variance decompositions and effect size expressions as their two-level counterparts.

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(Appendix follows)

Appendix

This appendix gives the solutions for the multilevel model parameters. The online supplemental materials provide details about their derivations. Given the predictor correlations, the intraclass correlations and variance values fully determine the variance–covariance matrices of the predictor variables (Φ_w , Φ_b , and Φ_p).

The coefficient solutions below require weight vectors described in the body text. To account for variational differences among the predictors, we compute rescaled weights by dividing each raw weight by the variable's standard deviation. For example, X_1^w 's rescaled weight is obtained by dividing its raw weight by its within-cluster standard deviation (i.e., the square root of the first diagonal element in Φ_w). Given the rescaled weights, the coefficient solutions are

$$\beta_w = \tilde{\omega}_w \sqrt{\frac{\text{var}(Y_{ij})R_w^2}{\tilde{\omega}_w' \Phi_w \tilde{\omega}_w}} \quad (\text{A1})$$

$$\beta_p = \tilde{\omega}_p \sqrt{\frac{\text{var}(Y_{ij})R_p^2}{\tilde{\omega}_p' \Phi_p \tilde{\omega}_p}} \quad (\text{A2})$$

$$\beta_b = \tilde{\omega}_b \sqrt{\frac{\text{var}(Y_{ij})R_b^2}{\tilde{\omega}_b' \Phi_b \tilde{\omega}_b}}, \quad (\text{A3})$$

where $\tilde{\omega}$ denotes a set of rescaled weights. When Level 1 predictors are grand mean centered, β_b requires an expression that partitions the between-cluster covariance matrix into submatrices for the Level 2 cluster means ($\Phi_{b(1)}$), pure Level 2 predictors ($\Phi_{b(2)}$), and covariances between the cluster means and pure Level 2 variables

($\Phi_{b(1,2)}$).

$$\beta_b = \tilde{\omega}_b \sqrt{\frac{\text{var}(Y_{ij})R_b^2}{\tilde{\omega}_b' (\Phi_{b(2)} - ((\Phi_{b(1)})^{-1} \Phi_{b(1,2)})' \Phi_{b(1,2)}) \tilde{\omega}_b}}. \quad (\text{A4})$$

The intercept is then obtained via subtraction from the arbitrary outcome mean.

$$\beta_0 = \mu_Y - \mu_w' \beta_w - \mu_p' \beta_p - \mu_b' \beta_b. \quad (\text{A5})$$

For a model with a single random coefficient, the slope variance is

$$\sigma_{b_1}^2 = \frac{\text{var}(Y_{ij})R_{rc}^2}{\text{var}(X_{1ij}^w)}. \quad (\text{A6})$$

The online supplemental materials describe the procedure for multiple random slopes. The residual between-cluster variance of the random intercepts is⁵

$$\sigma_{b_0}^2 = (\text{ICC}_Y) \text{var}(Y_{ij}) - \beta_b' \Phi_b \beta_b \quad (\text{A7})$$

and the off-diagonal element of Σ_b is a function of the random effect variances and correlation. Finally, the within-cluster residual variance is obtained by subtraction.

$$\sigma_\varepsilon^2 = (1 - \text{ICC}_Y - R_w^2 - R_{rc}^2) \text{var}(Y_{ij}) \quad (\text{A8})$$

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⁵ The solution for the random intercept variance is more complex when random slope predictors have non-zero means. See the online supplemental materials for details.